Continuity and Differentiability

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1 Continuity

A function f is continuous at $x = c$ if

$$\lim_{x \to c} f(x) = f(c).$$

Equivalently,

$$\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c).$$

The left and right limits are

$$\text{LHL} = \lim_{h \to 0^+} f(c - h), \qquad \text{RHL} = \lim_{h \to 0^+} f(c + h).$$

If f and g are continuous, so are $f+g$, $f-g$, fg, f/g (where $g \neq 0$), and compositions when defined.

2 Logarithm Rules

$$\begin{aligned}
\log(ab) &= \log a + \log b \\
\log(a/b) &= \log a - \log b \\
\log(a^x) &= x \log a \\
\log_e e &= 1 \\
\log_a a &= 1 \\
\log 1 &= 0 \\
\log 0 &\text{ is undefined} \\
\log_a x &= \frac{\log_b x}{\log_b a}.
\end{aligned}$$

Note: The logarithm base must be positive and not equal to 1, and the argument must be positive.

3 Important Limits

$$\begin{aligned}
\lim_{x \to a} \frac{x^n - a^n}{x - a} &= n a^{n-1} \\
\lim_{x \to 0} \frac{\sin x}{x} &= 1 \\
\lim_{x \to 0} \frac{1 - \cos x}{x} &= 0 \\
\lim_{x \to 0} \frac{\tan x}{x} &= 1 \\
\lim_{x \to 0} \frac{e^{ax} - 1}{ax} &= 1 \quad (a \neq 0) \\
\lim_{x \to 0} \frac{a^x - 1}{x} &= \ln a.
\end{aligned}$$

4 Differentiability

A function f is differentiable at c when

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

exists.

4.1 Derivative Rules

$$\begin{aligned} (u+v)' &= u' + v' \\ (uv)' &= u'v + uv' \\ \left(\frac{u}{v}\right)' &= \frac{vu' - uv'}{v^2}. \end{aligned}$$

Every differentiable function is continuous, but a continuous function need not be differentiable. Differentiability on an interval requires differentiability at each interior point.

5 Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b), there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

6 Rolle's Theorem

If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, there exists $c \in (a, b)$ with

$$f'(c) = 0.$$

7 Logarithmic Differentiation

For $y = u(x)^{v(x)}$ with $u(x) > 0$,

$$\ln y = v(x) \ln u(x), \qquad \frac{y'}{y} = v'(x) \ln u(x) + v(x) \frac{u'(x)}{u(x)}.$$

To check differentiability at c, compare the left and right derivatives:

$$\lim_{h \to 0^+} \frac{f(c) - f(c-h)}{h}, \qquad \lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h}.$$

8 Important Exponential Properties

$$e^{\ln a} = a \quad (a > 0), \qquad e^{b \ln a} = a^b.$$

9 Parametric Differentiation

If $x = f(t)$ and $y = g(t)$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{dx}{dt} \neq 0.$$

10 Higher Order Derivatives

$$\frac{dy}{dx} = f'(x), \qquad \frac{d^2y}{dx^2} = f''(x), \qquad \frac{d^3y}{dx^3} = f^{(3)}(x).$$